from scapy.all impor

```
from scapy.all import sniff, IP, TCP, UDP,
ICMP
from collections import Counter


----------------------------------
# Statistics
----------------------------------


total_packets = 0
total_bytes = 0

protocol_count = Counter()
source_count = Counter()
destination_count = Counter()


----------------------------------
# Packet Handler
----------------------------------
```

```
def packet_callback(packet):

    global total_packets, total_bytes

    if not packet.haslayer(IP):
        return

    ip = packet[IP]

    total_packets += 1
    total_bytes += len(packet)

source_count[ip.src] += 1
    destination_count[ip.dst] += 1
```

Identify protocol

```
    if packet.haslayer(TCP):
        protocol = "TCP"
        source_port = packet[TCP].sport
        destination_port = packet[TCP].dport

    elif packet.haslayer(UDP):
        protocol = "UDP"
        source_port = packet[UDP].sport
        destination_port = packet[UDP].dport
```

```
    elif packet.haslayer(ICMP):
        protocol = "ICMP"
        source_port = "-"
        destination_port = "-"

    else:
        protocol = "OTHER"
        source_port = "-"
        destination_port = "-"

    protocol_count[protocol] += 1

    print("-" * 70)
    print(f"Protocol    : {protocol}")
    print(f"Source      : {ip.src}:{source_port}")
    print(f"Destination  : {ip.dst}:
{destination_port}")
print(f"Packet Size  : {len(packet)} bytes")
```

Statistics

```
def show_statistics():

    print("\n")
    print("=" * 70)
    print("                    NETWORK STATISTICS")
    print("=" * 70)

    print(f"Total Packets : {total_packets}")
    print(f"Total Bytes   : {total_bytes}")

    print("\nProtocol Statistics:")

    for protocol, count in
protocol_count.most_common():
        print(f"{protocol:<10} : {count}")

    print("\nTop Source IPs:")

    for ip, count in
source_count.most_common(5):
        print(f"{ip:<20} : {count} packets")

    print("\nTop Destination IPs:")

    for ip, count in
destination_count.most_common(5):
```

```
print(f"{ip:<20} : {count} packets")
```

Filter Selection

```
print("=" * 70)
print("                PACKET SNIFFER")
print("=" * 70)

print("\nSelect a filter:")
print("1. All packets")
print("2. TCP")
print("3. UDP")
print("4. ICMP")

choice = input("\nEnter choice: ")


if choice == "1":
    packet_filter = "ip"

elif choice == "2":
```

```
    packet_filter = "tcp"

elif choice == "3":
    packet_filter = "udp"

elif choice == "4":
    packet_filter = "icmp"

else:
    print("Invalid choice.")
    exit()


print("\nStarting packet capture...")
print("Press CTRL+C to stop.\n")
```

Start Sniffer

```
try:

    sniff(
        filter=packet_filter,
```

```
        prn=packet_callback,
        store=False
    )

except KeyboardInterrupt:

    print("\nStopping capture...")

    show_statistics()
```